BOSARI
THERMAL MANAGEMENT

Product catalogue

Water coolers, heat exchangers and heat pipe systems for power electronics.

Products in this catalogue

- ATLANTIS® BAE6 – Water cooler
- ATLANTIS® BAE9 – Water cooler
- COLUMBUS 1000 – Heat exchanger for electronic cabinets
- U-COOL® 1200/50 – Heat pipe cooling system – Type 1200/50

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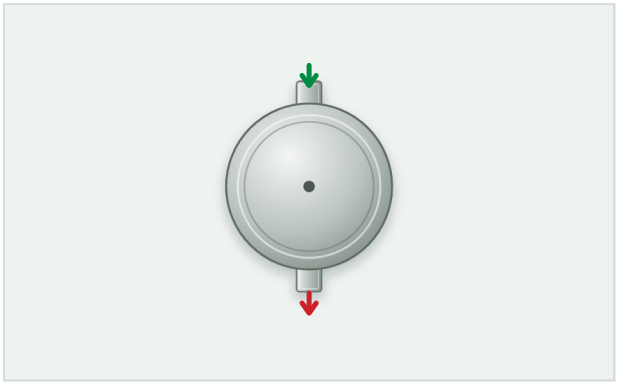
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BOSARI
THERMAL MANAGEMENT

DATA SHEET

ATLANTIS® BAE6

Water cooler



- **Up to 3,000 W (1,500 + 1,500)**
- **Contact area up to Ø 49 mm**
- **Weight approx. 160 g**

THERMAL	
Rth (single side) @6 l/min	**16 K/kW**
Rth (double side) @6 l/min	**10.5 K/kW**
Min. temp. of inlet fluid	**−25 °C**
Typ. temp. of inlet fluid	**+60 °C**
Max. operating temp.*	**+85 °C**

MECHANICAL AND ENVIRONMENTAL	
Clamping force up to	**60 kN**
Shocks and vibrations	**IEC 77 §16**
Cooling flow rate	**3 ÷ 9 l/min**
Max. pressure drop @6 l/min	**0.5 bar**
Typ. pressure drop @6 l/min	**0.3 bar**
Storage temperature	**−40 °C ÷ +85 °C**
Surface parallelism	**0.05 mm**
Surface planarity	**0.02 mm**
Surface roughness	**Ra 1.6**
Overall dimensions	**see drawing**
Weight	**160 g approx.**

* With pure water – usually depending on kind of cooling fluid and external circuit.

Description

During the course of recent years the development of electronic power components has made necessary to dissipate larger quantities of thermal heat at lower values of junction temperatures in order to increase their operating life.

The designers have become more and more familiar with alternative cooling systems, such as water cooling, that have shown the highest reliability in severe environmental conditions.

As an answer to high power dissipation problems, Bosari Thermal Management (BTM) has designed and developed a new family of aluminium water coolers, suitable to accept Thyristors, GTOs, and the recent IGBT components, dissipating up to 3.000W on both sides (1.500 + 1.500), with a contact area up to a diameter of 49mm (custom tailored up to 55mm).

This new cooler family can be easily adapted to the customer needs, by mounting the proper cooling fluid nipple (aluminium, titanium, etc.).

The basic concept of these coolers makes them really cost-effective, allowing a longer operating life to the components at a really reasonable price, thus permitting to adopt them in the most part of heavy industrial applications, welding, light and heavy electrical traction.

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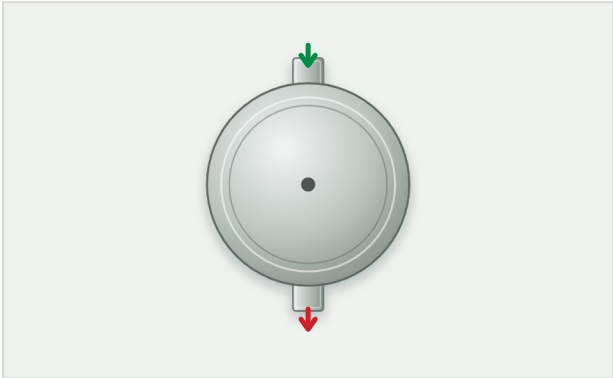
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DATA SHEET

ATLANTIS® BAE9

Water cooler



- **Up to 4,000 W (2,000 + 2,000)**
- **Contact area up to Ø 79 mm**
- **Weight approx. 330 g**

THERMAL	
Rth (single side) @6 l/min	**11 K/kW**
Rth (double side) @6 l/min	**7 K/kW**
Min. temp. of inlet fluid	**−25 °C**
Typ. temp. of inlet fluid	**+60 °C**
Max. operating temp.*	**+85 °C**

MECHANICAL AND ENVIRONMENTAL	
Clamping force up to	**90 kN**
Shocks and vibrations	**IEC 77 §16**
Cooling flow rate	**3 ÷ 9 l/min**
Max. pressure drop @6 l/min	**0.5 bar**
Typ. pressure drop @6 l/min	**0.3 bar**
Storage temperature	**−40 °C ÷ +85 °C**
Surface parallelism	**0.05 mm**
Surface planarity	**0.02 mm**
Surface roughness	**Ra 1.6**
Overall dimensions	**see drawing**
Weight	**330 g approx.**

* With pure water – usually depending on kind of cooling fluid and external circuit.

Description

During the course of recent years the development of electronic power components has made necessary to dissipate larger quantities of thermal heat at lower values of junction temperatures in order to increase their operating life.

The designers have become more and more familiar with alternative cooling systems, such as water cooling, that have shown the highest reliability in severe environmental conditions.

As an answer to high power dissipation problems, Bosari Thermal Management (BTM) has designed and developed a new family of aluminium water coolers, suitable to accept Thyristors, GTOs, and the recent IGBT components, dissipating up to 4.000W on both sides (2.000 + 2.000), with a contact area up to a diameter of 79mm (custom tailored up to 85mm).

This new cooler family can be easily adapted to the customer needs, by mounting the proper cooling fluid nipple (aluminium, titanium, etc.).

The basic concept of these coolers makes them really cost-effective, allowing a longer operating life to the components at a really reasonable price, thus permitting to adopt them in the most part of heavy industrial applications, welding, light and heavy electrical traction.

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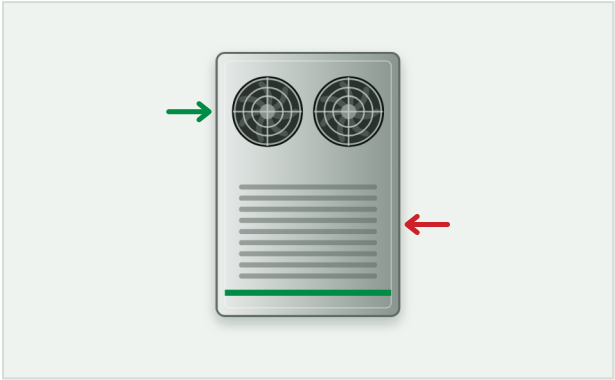
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DATA SHEET

COLUMBUS 1000

Heat exchanger for electronic cabinets



- **100 W/K cooling efficiency**
- **IP55 and NEMA 12**
- **Fans 115 V/60 Hz, 220 V/50 Hz or DC 12-24-48 V**

TECHNICAL CHARACTERISTICS	
Cooling efficiency	**100 W/K**
Dimensions	**440 × 1050 × 106 mm**
External case material	**Al 99.5%**
Exchanger core material	**Pure aluminium**
Operating temp. range	**−40 °C ÷ +100 °C**
Weight	**19 kg approx.**
Inlet fans	**2 × Ø 172 mm**
Outlet fans	**2 × Ø 172 mm**
Typ. air flow speed	**3.5 m/s**
Typ. noise @3.5 m/s	**58 dB(A)**
EMC compliances	**EN55022 · EN50082-2 · ETS300 342-2**

Description

The COLUMBUS family of passive heat exchangers has been specifically designed for electronic cabinets and environmental chambers. It features an innovative, extremely lightweight construction system, and an easy-to-mount chassis.

These highly efficient units, developed upon the concept of counterflow heat exchangers, have two separated hydraulic circuits, with no air exchange between the cabinet and the ambient.

They comply with IP55 and NEMA 12 requirements.

They can be mounted in any position, offering a great flexibility in designing devices, and they are maintenance free.

The COLUMBUS heat exchangers can be equipped with standard 115V/60Hz or 220V/50Hz fans, or with DC brushless fans (12-24-48V), all provided with finger guards.

The modules feature a thermal switch set at 45°C ± 5°C and NTC sensor for external temperature control. Custom-tailored control systems are available upon demand.

All the electrical connections are available on standard 3x2mm faston terminals.

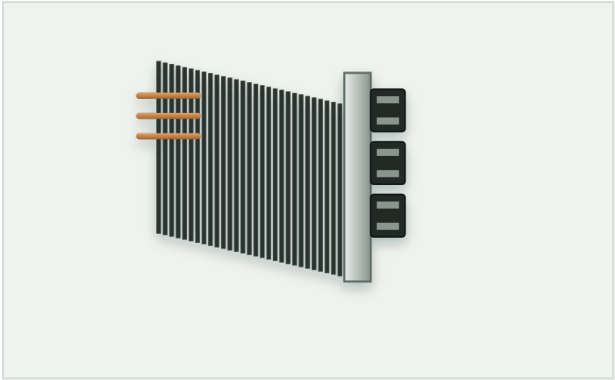
Materials and components used in the manufacturing of Bosari Thermal Management's COLUMBUS heat exchangers are carefully selected and quality controlled, to ensure the compliance with the requested specifications.

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U-COOL® 1200/50

Heat pipe cooling system – Type 1200/50



- **1,200 W by natural convection**
- **6 copper “U” heat pipes, 29 couples of fins**
- **Weight approx. 19 kg**

THERMAL	
Total thermal power	**1200 W**
Type of cooling	**Natural air convection**
Rth (heatsink baseplate – ambient air)	**< 50 K/kW**

MECHANICAL	
Plate surface planarity	**0.05 mm**
Plate surface roughness	**1.2 µm**
Plate finish	**Chemical white alodine 1000 and further black painting on pipe side**
Fins finish	**Black epoxy painting**
Weight	**19 kg approx.**

ENVIRONMENTAL	
Operating position	**Vertical**
Tilting angle	**± 8°**
Ambient temperature range	**−25 °C ÷ +80 °C**
Shock and vibrations	**As per IEC 77 §16**

Description

Description

This type of Heat Pipe Cooler has been designed to meet the general specification prepared by one of our customers, regarding the cooling of an IGBT inverter for application on railway vehicles, mounted underfloor with fins oriented in a parallel plane to the moving direction.

Operation

The heat produced by IGBT modules, mounted on a base plate, is carried, by means of 6 “U” shaped heat pipes, to a radiating section composed by 29 couples of fins mechanically housed on the heat pipes. Fins in turn dissipate the heat into the ambient, through a natural air convection.

The particular shape of the cooler also provides a very high efficiency with the forced air, caused by the movement of the train (assisted ventilation).

Construction

The sample unit presented here basically consists of:

1. A base plate made of Aluminum alloy, machined on one face to house the lower leg of the heat pipes by means of brazings or mechanical clamping, and on the other face to allow the mounting of IGBT modules and Diodes by means of tapped threaded holes. On this same surface a groove for housing a gasket can be provided.
2. A series of 6 heat pipes made of Copper, housed and soldered to the base plate by means of a special soldering alloy. Thermocarrying fluid can be demineralised water or fluorinert. Pipes are inclined of 10° over the horizontal line to allow an operation mode gravity assisted.
3. A dissipating section consisting of 29 couples of fins made of Aluminum alloy, each having dimensions of 420x135x1 mm. Fins are separated in order to provide a better flow of the cooling air. Fins have six holes each to match the pipes size and are inserted on the pipes with a mechanical interference.
4. A special closing aluminum fin, of 3 mm thickness, is provided to ensure a stiffening of the pipes structure and an additional fixing point to the train structure.